

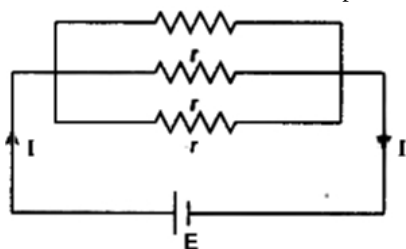
Solution

ELECTRICITY -07

Class 10 - Science

- In solid the **atoms** are very tightly packed with little space which makes the electrons to travel smoothly in the solid crystal similar to a **vacuum**. The electrons' **motion in a conductor** is different from the **empty space charges**. The electrons move with average drift speed when the steady current flows through the conductor.
- The electrons are not consumed in an electric circuit. The energy provided by the electricity company is used to move electrons through electric gadgets like an electric bulb, fan and engines. We pay to electricity board for providing the energy.
- The direction of flow of current in a circuit is opposite to the flow of electrons.
- Properties of Charge:
 - Electric charge can neither be created nor be destroyed but it can transfer from one body to another i.e., total electric charge in an isolated system is conserved.
 - Total charge on a body is equal to the algebraic sum of all the charges located on that body.

- Three resistors are connected in parallel. Let its equivalent resistance be $R \Omega$.



$$\therefore \frac{1}{R} = \frac{1}{r} + \frac{1}{r} + \frac{1}{r}$$

$$\frac{1}{R} = \frac{1+1+1}{r} = \frac{3}{r}$$

$$\Rightarrow R = \frac{r}{3}$$

By Ohm's law, $V = IR$

$$\Rightarrow \text{Current (I)} = \frac{E \times 3}{r} = \frac{3E}{r}$$

These resistances should be connected in parallel with the battery to obtain the maximum current.

- Electric circuit is a closed path through which the electrons flow readily. when a conductor is connected to a battery, the electrons move from negative terminal of the battery to the positive terminal. However, conventionally, the current is supposed to move from positive to negative.

Conventionally, the following symbols are used for different components:

Component	Symbol
An electric cell	or
Battery	
Open plug or switch	
Closed plug or switch	
Resistor	
Variable Resistor	
Galvanometer	
Ammeter	
Voltmeter	

- Conventional current is a direction opposite to that in which electronic current i.e., it is in direction opposite to that in which electrons flow. Electric current can be either negative or positive, but conventional current is always positive.
- Rate of flow of electric charge through any conductor is called electric current. Its SI unit is ampere(A).

Given, $I = 5A$, $t = 2 \text{ min} = 2 \times 60 \text{ s} = 120 \text{ s}$,

We know that,, $q = I \times t$

$$\Rightarrow q = 5 \times 120 = 600 \text{ C}$$

Thus, the amount of charge flowing through conductor is 600 C.

9. i. Negligible current will pass through the circuit because the voltmeter has a very high resistance.
ii. Ammeter will get damaged due to flow of large amount of current through it, because it has low resistance.

10.	Resistance	Resistivity (or Specific Resistance)
	The S.I. unit of resistance is Ohm (Ω).	The S.I. unit of resistivity is ohm-metre (Ω m).
	Resistance of a substance depends on its length and thickness.	Resistivity of a substance is independent of its length and thickness.

11. Both the wires will have same specific resistance, since they are both made of same material (i.e., copper) and there is no change in temperature.

12. For first wire

$$R_1 = \rho \frac{l}{A} = 4\Omega$$

Now for the second wire

$$R_2 = \rho \frac{l/2}{2A} = \frac{1}{4} \rho \frac{l}{A}$$

$$R_2 = \frac{1}{4} R_1$$

$$R_2 = 1\Omega$$

The resistance of the new wire is 1Ω

13. The resistivity of a given material is defined as the resistance of a conductor made of that material of unit length and unit cross-sectional area. It can be further defined as the resistance offered by a cube of that material of side 1 m, when current flows perpendicular to the opposite faces. Its S.I unit is ohm-metre (Ω m)

Using, $R = \rho \frac{l}{A}$, it can be noted that $R \propto l$ and $R \propto 1/A$

So, as the length of the wire is doubled, (area of cross section becomes half, if it is stretched to double its size). Hence, the resistance of the wire gets doubled.

From Ohm's Law, $V = IR$

If V remains unchanged, with resistance getting doubled, current becomes half of its original value.

14. It is given that resistance of the wire, $R = 4\Omega$ Thus, we have

$$R = 4 = \rho l / A, \text{ where } l = \text{length and } A = \text{cross-section of the wire.}$$

When a wire is doubled on it, its effective length becomes half and area of cross-section becomes doubled. Thus, $l' = \frac{l}{2}$ and $A' = 2A$

$$\text{Hence, the new resistance, } R_1 = \rho \frac{l/2}{2A}$$

$$\text{Therefore, } \frac{R_1}{R} = \frac{\rho \frac{l/2}{2A}}{\rho \frac{l}{A}} = \frac{1}{4}$$

$$\Rightarrow R_1 = \frac{R}{4}$$

$$= \frac{4\Omega}{4}$$

$$= 1\Omega$$

$\therefore$ The new resistance of the wire is 1Ω .

15. Resistance of wire $A(R_1) = \frac{\rho l}{A} = \frac{\rho l}{\pi r^2}$

$$\text{Resistance of wire } B(R_2) = \frac{\rho l'}{A'} = \frac{\rho 2l}{\pi (2r)^2} = \frac{\rho 2l}{4\pi r^2}$$

Total resistance in series

$$R = R_1 + R_2$$

$$R = \frac{\rho l}{\pi r^2} + \frac{\rho 2l}{4\pi r^2}$$

$$R = \frac{\rho l}{\pi r^2} \left(1 + \frac{1}{2}\right) = \frac{3\rho l}{2\pi r^2}$$

Ratio of the total resistance in series to the resistance of A =

$$\frac{R}{R_1} = \frac{\frac{3\rho l}{2\pi r^2}}{\frac{\rho l}{\pi r^2}}$$

$$\frac{R}{R_1} = \frac{3\rho l}{2\pi r^2} \times \frac{\pi r^2}{\rho l}$$

$$\frac{R}{R_1} = \frac{3}{2}$$

So, the required answer = 3:2

16. a. Factors affecting resistance of a conductor:

- i. Resistance is directly proportional to the length of the conductor

- ii. Resistance is inversely proportional to the area of cross-section of the conductor
- b. Metals are a good conductor of electricity because they have low resistivity and also have free electrons.
Glass is a bad conductor of electricity because it has high resistivity and have no free electrons.
- c. Alloys are commonly used in electrical heating devices because they have high resistivity and high melting point. They do not get oxidized (or burn) readily at high temperatures.
17. The resistance of a conductor is directly proportional to temperature ($R \propto t$). When a metallic conductor is heated, the atoms in the metal vibrate with greater amplitude and frequency. Due to increase in temperature, the thermal velocities of free electrons also increase. Therefore, the number of collisions between free electrons and atoms increases. This increases the opposition to the movement of electrons and hence the resistance of the conductor increases.
18. (a) Electrical resistance of a conductor may be considered as a measure of the opposition offered by it for flow of electric charge through it.
Mathematically, $R = V/I$. Its S.I. unit is ohm (Ω)
- (b) (i) $R = \frac{\rho L}{A}$ (1)
New length $L' = 2L$ and $A' = \frac{A}{2}$ (since volume remains same)
Therefore, $R' = \frac{\rho L'}{A'}$
 $R' = \frac{\rho 2L}{A/2}$ (2)
 $R' = 4 \left(\frac{\rho L}{A} \right)$
 $R' = 4R$ (from relation 1)
Therefore, resistance of the wire becomes 4 times its original resistance.
- (b) (ii) Since Resistivity does not depend on the dimensions of a wire. So it remains unchanged.
19. Resistance R of a conductor depends upon length and area A of conductor. It has been experimentally confirmed that:
 $R \propto L$, the length of the wire
 $R \propto \frac{1}{A}$, where A is the area cross-section of the wire
Combining above, we have: $R \propto \frac{L}{A}$ or $R = \rho \frac{L}{A}$
Where ρ is a constant called specific resistance or electric resistivity of the material of the conductor.
If $L = 1$, $A = 1$, $\rho = R$.
i.e. specific resistance (or electric resistivity) of a conductor is the resistance of a wire of unit length and a unit area cross-section.
It may be defined as:
Specific resistance of a conductor is the resistance of unit cube of the conductor.
Unit of ρ :
 $\rho = \frac{RA}{L} = \frac{\text{ohm m}^2}{\text{m}} = \text{ohm m}$
20. When given length of wire is doubled on itself, its new length $L' = \frac{L}{2}$ and new cross-section area $A' = 2A$. Hence, its new resistance.
 $R' = \frac{\rho L'}{A'} = \frac{\rho \left(\frac{L}{2} \right)}{(2A)} = \frac{1}{4} \frac{\rho L}{A} = \frac{R}{4}$
Thus, resistance is reduced to one-fourth of its original value.
21. $\therefore$ All of them are made of copper, the resistivity remains same. Let the resistivity be ρ
- i. $R_a = \rho \frac{L}{A}$
ii. $R_b = \rho \left(\frac{2L}{A/2} \right)$
 $= \rho \left(\frac{4L}{A} \right)$
 $= 4R_a$
iii. $R_c = \rho \left(\frac{L}{2(2A)} \right) = \frac{1}{4} \rho \left(\frac{L}{A} \right) = \frac{1}{4} R_a$
 $\therefore$ Conductor (b) has the highest resistance. It has the largest length and the smallest area of cross-section compared to other two. The following conclusion is also proved mathematically.
22. Resistance of a wire is inversely proportional to its area of cross-section (or thickness). Hence, a connection wire should be thick to reduce its resistance.
23. The current flows more easily through a thick wire as compared to thin wire of the same material, when connected to the same source. Resistance is inversely proportional to the area of cross section of the wire. It is due to this reason that resistance increases with decrease in thickness.

24. The resistance of a wire (R) depends on the following factors:
- Resistance is directly proportional to the length of the wire. $R \propto l$ (length of the conductor)
 - Resistance is inversely proportional to the area of cross-section of the wire. $R \propto \frac{1}{A}$ (area of cross - section)
 - $R \propto t$ (temperature)
 - R depends on the nature of material.
25. If 'n' electrons pass through the cross-section of a conductor in time Y, the total charge 'Q' passing through the conductor is:
- $Q = ne$ (e is the charge on an electron = 1.6×10^{-19} C)
- The current **I** in the conductor is:
- $$I = \frac{Q}{t} = \frac{ne}{t}$$
26. i. Resistance depends on the length and area of a substance.
- Resistivity depends on the nature and temperature of the substance.
- ii. Given:
- $$\rho = 1.63 \times 10^{-8} \Omega - m$$
- $$A = 10.3 \times 10^{-4} \text{ cm}^2 = 10.3 \times 10^{-4} \times 10^{-4} \text{ m}^2$$
- $$R = 20 \Omega, l = ?$$
- We know that, Resistance, $R = \frac{\rho l}{A}$
- $$\Rightarrow l = \frac{RA}{\rho} = \frac{20 \times 10.3 \times 10^{-4} \times 10^{-4} \text{ m}^2}{1.63 \times 10^{-8} \Omega - m} = \frac{20 \times 10.3 \times 10^{-8}}{1.63 \times 10^{-8}} = 126.38 \text{ m}$$
27. Copper contains a large number of free electrons, hence is a good conductor of electric current whereas rubber has negligible number of free electrons and is an insulator.
28. The cable carrying 10 A current will be of thicker wire because to carry a heavy current, the resistance of wire should be low, hence its area of cross-section should be large.
29. $d = 0.5 \text{ mm} = 0.5 \times 10^{-3} \text{ m} = 5 \times 10^{-4} \text{ m}$; $\rho = 1.6 \times 10^{-8} \Omega \text{ m}$, $l = ?$; $R = 10 \Omega$
- $$l = \frac{R \pi d^2}{4 \rho} = \frac{10}{4} \times \frac{22}{7} \times \frac{(5 \times 10^{-4})^2}{1.6 \times 10^{-8}}$$
- $$R = \frac{\rho l}{A} = \frac{\rho l}{\frac{\pi d^2}{4}} = \frac{4 \rho l}{\pi d^2}$$
- $$l = \frac{220}{28} \times \frac{25}{1.6} = 122.77 \text{ m}$$
- If the diameter of wire is doubled, resistance will become $\frac{1}{4}$ th of the original (since $R \propto \frac{1}{A}$ or $R \propto \frac{1}{d^2}$).
- Decrease of resistance will be $10 - \frac{10}{4} = 7.5 \Omega$
30. Electric fuse works on the principle of heating effect of electric current. When a large amount of current flows through the circuit, temperature of the wire increases and fuse wire gets melted due to its low melting point. This in turn prevents the flow of current in the other circuits of the house and thus save electrical appliances from any damage. Under normal conditions, the fuse wire is part of the circuitry contributing to a complete loop for charges to flow through it.
31. The heating element of the heater is made up of alloy which has very high resistance compared to the cord of an electric heater. Since, the heat produced is directly proportional to the resistance, it becomes too hot and glows red due to heating effect of the current. But the resistance of the cord which is usually of copper or aluminium is very low so it does not glow.
32. The filament of electric bulb is made up of material like tungsten which has high resistance, whereas the line wires are of negligible resistance. Since the amount of heat generated is proportional to the resistance ($H \propto R$), the filament generates much more heat and it starts glowing.
33. Cord of heater and electric heater are joined in series and carry same current when joined to voltage source. As resistance of cord is extremely small as compared to that of heater element, hence, heat produced $H = I^2 R t$ is extremely small in cord but much larger in heater element. So, the heating element begins to glow but cord does not glow.
34. For finding power, we need to first calculate the resistance in the circuit.
- $$R = \frac{V}{I} = \frac{4.5 \text{ V}}{3 \text{ A}} = 1.5 \Omega$$
- Now, $P = I^2 R = (3 \text{ A})^2 \times 1.5 \Omega = 13.5 \text{ W}$
35. Alloys are used in electrical heating devices rather than pure metals because they have higher resistivity as compared to pure metals. Hence, more heat is produced. Moreover, alloy does not burn (or oxidize readily at higher temperature).
36. Heating element of an electric iron is made up of nichrome wire because of the following reasons:
- It has high resistance and resistivity because of which more heat is produced due to the passage of current.
 - Melting point is high.
 - It does not oxidize (or burn) easily at high temperature. It forms an adherent layer of chromium oxide when it is heated for the first time and thus it is also resistant to corrosion.

37. a. The mathematical expression of the Joules Law of heating is: $H = I^2 R t$
Here, H is a heating effect, I is the current flowing through the device and t is the time taken.
- b. Given:
Amount of charge transferred = 96000 C
Time taken = 2hrs = $2 \times 60 \times 60$ sec = 7200 sec
Potential difference = 40 V
Heat generated = $V \times I \times t$
and we know that; $I = \frac{Q}{t}$
So, $H = VQ$
 $= 40 \times 96000$
 $= 3.84 \times 10^6$ J
38. The coiled wire of tungsten increases the surface area of the wire in very less space so as to emit more light and helps in glowing with more intensity.
39. a. Electrical energy gets converted into heat energy in an electric heater.
b. Electrical energy changes into mechanical energy in an electric motor.
c. Electrical energy gets converted into sound energy in a loudspeaker.
d. Electrical energy changes into chemical energy during electrolysis.
40. a. According to Joule's law of heating:
(i) The heat produced in the resistor is directly proportional to the square of the current
(ii) The heat produced in the resistor is directly proportional to the resistance and
(iii) The heat produced in the resistor is directly proportional to the time for which the current flows through the resistor.
Mathematically,
 $Q = I^2 R t$
where, Q = Heat produced, I = current, R = Resistance and t = time
- b. Current flowing in the first bulb, $I_1 = \frac{P_1}{V} = \frac{100}{220} = \frac{5}{11} = 0.45$ A
Current flowing in second bulb, $I_2 = \frac{P_2}{V} = \frac{60}{220} = \frac{3}{11}$ A = 0.27 A
So, line current $I = I_1 + I_2 = 0.45 + 0.27 = 0.72$ A
41. Lead-tin alloy is used for fuse wires because it has low melting point.
42. The Joule's Law of Heating states that the heat produced in a resistor is directly proportional to the square of current for a given resistor, resistance for a given current, and time for which the current flows through the resistor.
This can be expressed by following equation:
 $H = I^2 R t$
Here; I is electric current, R is resistance, t is time and H is heat produced.
- Experiment to Demonstrate Joule's Law of heating**
- Take a water heating immersion rod and connect to a socket and regulator.
 - Keep the pointer of regulator on minimum and count the time taken by immersion rod to heat a certain amount of water.
 - Increase the pointer of regulator to next level. Count the time taken by immersion rod to heat the same amount of water.
 - Repeat above step for further readings.
- Observation:** It is seen that with increased amount of electric current, less time is required. This shows Joule's Law of Heating.
- Application:** Electric toaster, oven, electric kettle, electric heater, induction cooker and geyser etc.
43. Here, Power (P) = 18 W and Resistance (R) = 2Ω
 $P = I^2 R$
or, $18W = I^2 \times 2\Omega$
or, $I^2 = \frac{18W}{2\Omega} = 9$
or, $I = 3$ A
3 A is the maximum current that can be withstood by each resistor of 2Ω .
44. (ii) Ammeter A shows a reading of 3A. This means each of the A_1 , A_2 and A_3 show 1A reading. When the bulb B_2 gets fused, no current flows through this bulb. So, all the current is equally divided between remaining two bulbs. So, ammeter A_1 and A_2 will show $-\frac{3}{2} = -1.5$ A current each. Ammeter A_3 will show zero current.

45. $H = 100 \text{ J}$, $R = 4\Omega$, $t = 1 \text{ s}$, $V = ?$

From eq. $H = I^2 R t$ we have the current through the resistor as

$$I = \sqrt{\left(\frac{H}{Rt}\right)}$$

$$= \sqrt{\left[\left(\frac{100\text{J}}{4\Omega \times 1\text{s}}\right)\right]}$$

$$= 5\text{A}$$

Thus the potential difference across the resistor, V eq. $V = IR$ is

$$V = IR$$

$$= 5\text{A} \times 4\Omega$$

$$= 20 \text{ V}$$

46. The resistivity of an alloy is generally higher than that of its constituent metals. Alloys do not oxidise (burn) readily at higher temperatures. Therefore, conductors of electric heating devices, such as toasters and electric irons, are made up of an alloy rather than pure metal.

47. Two features of the material used as an element of an electric iron are:

- i. It should have a low melting point.
- ii. It should have very high resistance.

48. Let resistance of each wire be R

In series, resistance is therefore $= 2R$

Heat produced in series arrangement, $H_S = \frac{V^2}{2R}(t)$

In parallel total resistance $= \frac{R}{2}$

Heat produced in parallel arrangement, $H_P = \left(\frac{V^2}{\frac{R}{2}}\right)(t) = \frac{2V^2}{R}t$

$\therefore H_P = 4H_S$ [$\because H_S = \frac{V^2}{2R}(t)$]

$\therefore$ The required ratio is 1:4

$\therefore$ Option (c) is correct.

49. We know that the conductors offer some resistance to the flow of current. Resistance in a conductor or electric circuit is analogous to friction. Just as we have to do work against friction. Let us calculate the amount of work done in carrying charge Q through a potential difference V . We know that the potential difference between two points of a conductor is one volt if one joule of work is done in carrying one coulomb of charge from one point to the other. Therefore, if I is the current in ampere flowing for t second, the quantity of charge $Q = It$.

The work done in carrying a charge $Q = (It)$, coulomb against a potential difference of V volt $= VIt$ joule.

$\therefore$ work done in carrying Q units of charge against a P.D. of V volt is

$$W = V (It) = VIt \text{ joule}$$

This is the energy spent in overcoming the resistance offered by the conductor. Whole of this energy is converted into heat.

Heat produced $H = VIt$ joule

Now $V = IR$ $H = (IR)It$ joule or $H = I^2 R t$ joule

Again $I = \frac{V}{R}$

$\therefore H = \frac{V^2}{R^2} R t$ Joule or $H = \frac{V^2}{R} t$ joule

50. The resistance of a heater coil is less than that of electric bulb filament. When the heater is switched on in parallel, more current start flowing through the heater coil and current through the bulb filament decreases, making it dim.

After some time, when the heater coil becomes hot its resistance increases ($H \propto R$). As a result, current through the heater coil decreases and hence, the current through the bulb filament increases and thus dimness of the bulb decreases.

51. The three important effects produced by electric current are

- a. Heating effect,
- b. Magnetic effect,
- c. Chemical effect.

52. Rate at which heat is developed in the heater is its power.

$$P = I^2 R = (15)^2 (8) = 225 \times 8 \text{ or } P = 1800\text{W or } \text{Js}^{-1}$$

1,800 J of heat is developed per second.

53. i. Radhika put off the main switch quickly to save her mother.

ii. Red wire which is at a high potential of 220V was touching the metal case of electric kettle.

iii. The values displayed by Radhika are:

- a. Concern for her mother.
- b. Presence of mind.
- c. Knowledge of household wiring and daily life activities.

54. Work done to transfer Q coulomb of charge through potential difference $\Omega = VQ$ joule.

$$\Omega = 50 \times 96,000 = 48,00,000 \text{ J}$$

$$\Omega = 4.8 \times 10^6 \text{ K}$$

Which is equal to heat generated.

55. When the temperature of the heater becomes greater than the temperature of the surroundings, some of the heat is lost to the surroundings in the form of thermal radiations and through convection. After some time, the rate at which heat is being produced becomes equal to the rate at which heat is lost. Hence, the temperature of the element becomes constant. This happens because an equilibrium or steady state is reached.

56. Resistance of heater coil,

$$\begin{aligned} R &= \frac{V^2}{P} \\ &= \frac{(200)^2}{100} \\ &= \frac{200 \times 200}{100} \\ &= 400\Omega \end{aligned}$$

As the coil is cut into two identical parts, resistance of each part = $\frac{400}{2} = 200\Omega$

When connected in parallel, the net resistance is given by

$$\begin{aligned} \frac{1}{R_p} &= \frac{1}{200} + \frac{1}{200} \\ \Rightarrow \frac{1}{R_p} &= \frac{2}{200} \\ \therefore R_p &= 100\Omega \end{aligned}$$

Energy liberated = $P \times t$

$$\begin{aligned} &= \frac{V^2}{R} \times t = \frac{(200)^2}{100} \times 1 \\ &= 400\text{J} \end{aligned}$$

$\therefore$ Energy liberated per second in the new combination is 400J

57. We know that, $P = I^2 R$

Current after increased by 100% = $I + \frac{100I}{100} = 2I$

$$P' = (2I)^2 R = 4I^2 R$$

Percentage increase in power dissipation is

$$\begin{aligned} &= \frac{P' - P}{P} \times 100 \\ &= \frac{4I^2 R - I^2 R}{I^2 R} \times 100 \\ &= \frac{I^2 R(4-1)}{I^2 R} \times 100 = 3 \times 100 = 300\% \end{aligned}$$

58. Power is the rate of doing work. The electric power of an appliance is its rate of consumption of electric energy.

The power of an electric appliance is 1 Ω (watt) if it consumes 1J (joule) of energy in 1 second.

If I ampere of the current flows for t second through a coil whose ends are maintained at a potential difference of V volt, then the energy consumed or the work done.

$$\Omega = VIt \text{ joule}$$

$$\text{Power} = \frac{W}{t} = \frac{VIt \text{ joule}}{t \text{ second}}$$

$$\text{or } P = V \times I \text{ joule/second} = VI \text{ watt}$$

$$P(\text{watt}) = V(\text{volt}) \times I(\text{ampere}) \text{ or watt} = \text{volt} \times \text{ampere}$$

Again we know that $V = IR$

$$P = VI = (IR)I = I^2 R \text{ or } P = I^2 R$$

Also, $I = V/R$, therefore

$$P = I^2 R = \left(\frac{V}{R}\right)^2 R \text{ or } P = \frac{V^2}{R}$$

Bigger unit of power usually k Ω (kilowatt)

$$1\text{k}\Omega = 1,000 \Omega$$

$$P(\text{in kW}) = \frac{\text{Watt}}{1000} = \frac{V(\text{volt}) \times I(\text{ampere})}{1000}$$

59. We know that

$$P = \frac{V^2}{R}$$

$$\text{Therefore, } R = \frac{V^2}{P}$$

Resistance of 1st lamp,

$$R_1 = \frac{V^2}{P} = \frac{220 \times 220}{100}$$

$$= 484 \Omega$$

Resistance of 2nd lamp,

$$R_2 = \frac{220 \times 220}{60}$$

$$= \frac{2420}{3} \Omega$$

Since, two lamps are connected in parallel, so its equivalent resistance is given by

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{484} + \frac{3}{2420} = \frac{8}{2420}$$

$$R = \frac{2420}{8} \Omega$$

By Ohm's Law, current drawn from the mains:

$$I = \frac{V}{R} = \frac{220 \times 8}{2420}$$

$$= 0.73 \text{ A}$$

$\therefore$ The current drawn from the mains is 0.73 A

$$60. \text{ Energy consumed} = \frac{\text{Watt} \times \text{hour}}{1000} = \frac{50 \times (2 \times 30)}{1000} = \frac{3000}{1000} = 3 \text{ unit or } 3 \text{ kWh}$$

$$61. \text{ i. } P = \frac{V \times V}{R}$$

$$R = \frac{V \times V}{P}$$

$$R = \frac{220 \times 220}{1100}$$

$$R = \frac{48400}{1100}$$

$$R = 44 \text{ ohm}$$

$$\text{ii. } \frac{V}{I} = R$$

$$I = \frac{V}{R}$$

$$I = \frac{220}{44}$$

$$I = 5 \text{ A}$$

$$62. \text{ Energy consumed } E_1 = P_1 t_1 = 250 \text{ W} \times 1 \text{ h} = 250 \frac{\text{J}}{\text{s}} \times 3600 \text{ s}$$

$$E_1 = 900000 \text{ J}$$

$$E_2 = P_2 t_2 = 12000 \text{ W} \times 10 \text{ min}$$

$$E_2 = 1200 \frac{\text{J}}{\text{s}} \times 600 \text{ s}$$

$$E_2 = 720000 \text{ J}$$

$\therefore$ 250 Ω TV set consumes more energy.

63. Here, Power rating, $P = 1000 \text{ W}$

Voltage rating, $V = 220 \text{ V}$

$$\text{i. Using the relation, } P = \frac{V^2}{R}$$

Resistance of element when in use,

$$R = \frac{V^2}{P} = \frac{220 \times 220}{1000}$$

$$\text{ii. Using the relation, } P = VI$$

$$\text{Safe Current, } I = \frac{P}{V} = \frac{1000}{220}$$

$$= 4.55 \text{ A}$$

64. Wattage of colour T.V. = 60 W

$$\text{Number of hours for which colour T.V. is on during September} = 2 \times 2\frac{1}{2} \times 30 = 150 \text{ h}$$

$$\text{Energy consumed} = \frac{\text{watt} \times \text{hour}}{1000} = \frac{60 \times 150}{1000} = 9 \text{ kWh}$$

$$\text{Cost of seeing 2 movies per day for 30 days} = 9 \times 4 = \text{Rs. } 36 \text{ only.}$$

65. (i) $V = 6 \text{ V}$

$$R_1 = 1 \Omega$$

$$R_2 = 2 \Omega$$

$$\text{Total resistance, } R = R_1 + R_2$$

$$= 1 + 2 = 3\Omega$$

$$I = \frac{V}{R}$$

$$= \frac{6}{3}$$

$$= 2A$$

Therefore, power (P_1) = I^2R

$$= (2)^2 \times 2$$

$$= 8W$$

$$(ii) V = 4V, R_1 = 12\Omega, R_2 = 2\Omega$$

$\therefore$ Potential difference is constant in a parallel circuit. Hence power dissipated across the 2Ω resistor is directly given as:

$$P_2 = \frac{V^2}{R}$$

$$= \frac{4 \times 4}{2}$$

$$= 8W$$

$$\text{Ratio } \frac{P_1}{P_2} = \frac{8}{8} = 1:1$$

As an interesting observation might be that power dissipated in parallel circuit does not depend on the resistor connected in parallel, while in a series circuit, as the current flow is determined by all the resistors, power dissipation is dependent on the other resistors in the circuit.

66. (i) When a 2Ω resistor is joined to a 6 V battery in series with 1Ω and 2Ω resistors. Total resistance (R_s) = $2 + 1 + 2 = 5\Omega$.

$$\therefore \text{Current } (I_1) = 6V/5\Omega = 1.2 A$$

$$\therefore \text{Power used in } 2\Omega \text{ resistor, } P_1 = I_1^2 R = (1.2)^2 \times 2 = 2.88 W$$

(ii) When 2Ω resistor is joined to a 4 V battery in parallel with 12Ω resistor and 2Ω resistors, the current flowing in 2Ω is independent of the other resistors.

$$\therefore \text{Current flowing through } 2\Omega \text{ resistor, } I_2 = 4 V / 2\Omega = 2 A$$

$$\text{Power used in } 2\Omega \text{ resistor, } P_2 = I_2^2 R = (2)^2 \times 2 = 8 W$$

$$\therefore \text{The required ratio, } P_1 / P_2 = 2.88/8 = 0.36 : 1$$

67. In parallel circuit, potential difference does not get divided. Hence, glow of other bulbs will not be affected when bulb B_1 get fused.

68. Rate at which energy is delivered by the current is called power. It is measured in watt or in kilowatt.

$$P = VI$$

Power in watt = V in volt $\times$ I in ampere.

If $V = 1$ volt; $I = 1$ ampere; then $P = 1$ watt

$\therefore$ Power consumed by a device is 1 W, if 1A of current flows across a potential difference of 1 V.

69. Here $V = 220$ volts; $R = 440\Omega$

$$\text{Now } I = \frac{V}{R} = \frac{220}{440} = 0.5 A$$

$$\text{Heat energy produced in 30s} = \frac{V^2}{R} T = \frac{(220)^2 \times 30}{440} = 3300 J$$

70.	Electric Energy	Electric Power
	Electrical energy consumed by an electrical appliance is given by the product of its power rating and time for which it is used.	It is the rate at which electrical energy is consumed.
	It is measured in kWh.	It is measured in watt or kilo-watt.

71. Let current through each bulb be I.

$$P = VI, 10 = 220 I$$

$$I = \frac{1}{22} A$$

Let n such bulbs be connected in series.

$$\text{Current through n bulbs} = 5A$$

$$n (\text{current in 1 bulb}) = 5$$

$$n \frac{1}{22} = 5$$

$$n = 110$$

110 such bulbs can be lighted within allowable limit of 5A.

72. Power = $P = VI = 220 \times 5 = 1,100$ watt

Energy consumed in 2 h ($2 \times 60 \times 60$ s) $E = Pt$

$E = (1,100) (2 \times 60 \times 60) = 7,920,000$ J

$E = 7.92 \times 10^6$ J

73. Here, $V = 220$ volt, $P = 1$ kW = 1000 W, $t = 2$ minute = $2 \times 60 = 120$ s

i. Resistance of the heating element,

$$R = \frac{V^2}{P} = \frac{(220)^2}{1000} = 48.4 \Omega$$

ii. Current through the element, $I = \frac{P}{V} = \frac{1000}{220} = 4.54$ A

iii. Heat produced in 2 minutes = $P \times t$

$$= 1000 \times 120 = 1.2 \times 10^5 \text{ J}$$

iv. If line voltage falls to 200 V, the power consumed is,

$$P = \frac{V^2}{R} = \frac{(200)^2}{48.4} \\ = 826.44 \text{ W}$$

74. Energy consumed (in kWh) = power (in kW) $\times$ time (h) = $2.2 \text{ kW} \times 3\text{h} = 6.6 \text{ kWh}$

Power = $2.2 \text{ kW} = 2.2 \times 1,000 \text{ W} = 2,200 \text{ W}$

But Power = Voltage $\times$ Current

$$2,200 = 220 \times I$$

$$I = \frac{2200}{220} = 10 \text{ A}$$

75. The circuit diagram showing the electric fuse and earthing as safety devices is as follows:

